

Packaging Investigation Brief

Fictional training source — no customer data

Investigation ID: QI-26-017
Product family: Velvet Bloom 50 mL jar
Process: Line 4 induction sealing
Review period: August 18–September 18, 2026
Status: Open; product on hold

Executive summary

Routine in-process inspection identified intermittent low seal-strength readings during extended Line 4 runs. No released product is associated with this fictional event. Initial evidence does not support supplier film thickness as the primary cause. The leading hypothesis is temperature-control drift after three hours of continuous operation, but causality has not been confirmed.

What happened

- On August 18, three of 80 sampled jars produced seal-strength readings below the internal alert level during hour four of a production run.
- The affected material was placed on hold and investigation QI-26-017 was opened.
- Review of the previous six runs found no reading below the release limit, but two runs showed increasing variation after hour three.
- The temperature sensor was replaced and calibrated on September 3.
- A four-hour engineering run on September 12 remained within the provisional acceptance range but showed greater variation in hour four.

Evidence reviewed

Evidence	Result	Interpretation
Supplier certificates	Met specification	No certificate exception identified
Independent film-thickness test	Met specification for all retained rolls	Supplier material reduced as primary suspect
Temperature-sensor calibration	Pre-replacement drift observed; replacement passed	Supports equipment-control hypothesis
Four-hour engineering run	No release-limit failure; variability increased after hour three	Signal requires a longer confirmation run
Prior six-run trend	Two runs showed late-run variability	Pattern is possible but not yet conclusive

Risk assessment

The product remains on hold. No consumer exposure occurred in this fictional scenario. The current risk is operational: an unresolved late-run process-control issue could recur if Line 4 returns to normal production without confirmation. Existing hold controls reduce immediate product risk.

Working root-cause hypothesis

Temperature-control drift during extended operation may reduce sealing consistency after approximately three hours. This is a hypothesis, not a confirmed root cause. Supplier film thickness has not been identified as causal based on the evidence reviewed.

Actions and owners

Action	Owner	Due date	Status
Approve eight-hour run sampling plan	Lena Ortiz, Validation	September 22	Open
Complete eight-hour confirmation run	Engineering	September 24	Scheduled
Issue time-series trend analysis	Data analyst	September 28	Not started
Update investigation report	Mira Patel, Quality	September 30	In progress
Confirm disposition recommendation	Quality review board	After evidence review	Not started

Decision needed from the review group

Approve the proposed eight-hour confirmation run and sampling plan. Do not approve investigation closure, final root cause, product disposition, or return to routine production until the new evidence has been reviewed.

Known limitations

- The event set is small.
- The four-hour engineering run did not reproduce a release-limit failure.
- The relationship between sensor drift and seal strength is correlational at this stage.
- No final root cause or closure date has been approved.

Presentation guidance

When turning this brief into slides, distinguish confirmed facts from hypotheses. Keep the decision request explicit. Do not state that the sensor replacement solved the issue, and do not invent missing dates, test results, or approvals.